

Literature Review

SOC Automation Lab for Threat Detection and Incident Response

Introduction

Security Operations Centers (SOC) play a critical role in protecting modern organizations from cyber threats. A SOC is responsible for monitoring systems, detecting malicious activity, and responding to security incidents in real time. With the increasing number of cyber attacks, organizations rely on centralized monitoring platforms and automated response mechanisms to quickly detect and mitigate threats.

SIEM and Log Analysis

Security Information and Event Management (SIEM) platforms collect and analyze logs from different sources such as endpoints, servers, and network devices. SIEM solutions correlate events and identify suspicious patterns that may indicate malicious activity. Centralized log analysis allows security teams to detect attacks that would otherwise be difficult to identify when logs are scattered across multiple systems.

Endpoint Monitoring

Endpoint monitoring tools provide visibility into system behavior by tracking processes, file changes, registry modifications, and network connections. Monitoring endpoint activity is essential for detecting malware execution, privilege escalation, and lateral movement within a network environment.

Incident Response Platforms

Incident response platforms enable SOC analysts to investigate security alerts and manage incidents effectively. These platforms allow analysts to track investigation steps, correlate indicators of compromise, and document response procedures. Incident management tools improve collaboration and ensure that security incidents are handled in a structured and repeatable manner.

Threat Intelligence Integration

Threat intelligence services provide information about known malicious IP addresses, domains, and file hashes. Integrating threat intelligence into a SOC environment allows alerts to be enriched with additional context, helping analysts quickly determine whether an event is benign or malicious.

Attack Simulation and Adversary Emulation

To properly test detection capabilities, SOC environments often simulate real attacks. Adversary emulation tools allow researchers and security teams to replicate attacker behavior in a controlled environment. This project uses MITRE Caldera to simulate cyber attacks based on the MITRE ATT&CK framework. Caldera enables the execution of realistic attack techniques such as lateral movement, privilege escalation, and command execution. By simulating these attacks, the SOC lab can validate detection rules and improve defensive strategies.